

VOTEOMETER – the App That Helps You Choose Your Candidate

By Larry Hodges • larry@larrytt.com • Contact me if interested in developing this.

What follows is the initial idea, math, and development of a political app (for phones and webpages) that could be highly useful for a large number of people – perhaps going viral – in choosing in a primary election between the candidate you think would be the best president, and the most electable one. An early version has been created and is online, with link below. It needs to be further developed either by a professional site devoted to politics, by an ambitious entrepreneur, or perhaps college students as a college project and/or for profit (by developing and selling it to a professional site, or on its own as a viral app). Whoever develops it would need strong math and programming skills (either individually or as a collaboration). Once developed, they could either market the app on their own if they know how, or sell it to a major political or news site, such as Politico, 538, New York Times, Washington Post, CNN, MSNBC, NBC, ABC, CBS, NPR, AP, Pew Research, Real Clear Politics, and so on.

So, what is **Voteometer**? Every four years Americans face a quandary in presidential primary elections. Should they vote for who they think would make the best president (and risk getting stuck with the opposing party's candidate), or the one from their party who might not be their first choice but is more electable in the general election? What if you really would like to vote for the most appealing candidate, but are appalled at the possibility that in so doing, we may end up with the devil from the other side? How do you balance these off?

Until now, voters had to base this on their gut feeling. However, with Voteometer, all you have to do is rate the top candidates in each party in terms of how good *you* think they'd be as president, and *your own* judgement of their probability of winning head-to-head against the other party's candidates, and instantly get your answer on who you should vote for, based on *your own judgements*. (There should be a button that allows the user to use professional judgements for the head-to-head probabilities, if they choose.)

How does it do this? It takes into account how good or bad *you* judge the top candidates from each party would be as president, and the probability of each of them winning against each of the candidates from the other party (which you can judge on your own or use professional opinions). Once you've given these inputs, Voteometer calculates the odds of each candidate winning, based on your inputs, then multiplies the odds of each candidate winning times the value you gave the candidate, giving each candidate a “Power Number.” You then vote for the candidate from your party with the highest Power Number. (See examples on next page.) The numbers you put in may be subjective, but once inputted, it's 100% objective and accurate. The software calculates a “Power Number” for your candidates, and you vote for whoever has the higher number.

More specifically, the software asks whether you are a Democrat or Republican (Independents and other parties can be added later); has you rate each of the candidates on a scale from 10 (“God”) to -10 (“Satan”). For perspective, for a Democrat, an average Democratic candidate might be a 5, while an average Republican candidate might be a -5, and vice versa, while a purely average moderate candidate in the middle of the two parties would be a 0. The software then asks you to judge the probability each candidate has in winning a general election head-to-head against each of the opposing party candidates. (There should be an optional button that allows you to use “expert” opinions on this.)

I could continue working on it myself, but I'd rather bring in someone to develop it further at this point. I am not asking for any recompense for this, as long as I'm credited in some way as the initial developer and/or co-creator. Once someone takes it over, they have control of it and its development – I'm at most an occasional adviser. (Note – I have a bachelor's in math from long ago, but am basically retired.)

Before going further, why not give the initial version a try? For test purposes, two Democratic and Republican candidates are hard-wired in – Biden, Sanders, Trump, and DeSantis. This simplified version was put together by an associate of a friend of mine (Roberto J Miglietti), and serves primarily to show how the system works.

<https://alternatura.com/pave/>

UserID: testuser

Password: Summer2023

Now that you've seen this initial version in action, how does the math work? Let D1 and D2 be the top two Democratic candidates. Let R1 and R2 be the top two Republican candidates. Here are the questions you would receive.

Questions and Variables

- Are you a Democrat or Republican? [If Democrat, skip 6. If Republican, skip 7 You can skip these because you are deciding who to vote for, so the head-to-head probability in a two-way race doesn't matter. If the app is developed for three or more candidates, then the probability between the candidates would be needed.]
- On a scale of -10 to 10, how would you rate D1 as president? $D1 \text{ Presidential} = D1P$
- On a scale of -10 to 10, how would you rate D2 as president? $D2 \text{ Presidential} = D2P$
- On a scale of -10 to 10, how would you rate R1 as president? $R1 \text{ Presidential} = R1P$
- On a scale of -10 to 10, how would you rate R2 as president? $R2 \text{ Presidential} = R2P$
- In D1 vs. D2, what is the probability that D1 wins primary? $D1 \text{ Electability} = D1W$
 $D2 \text{ Electability} = D2W = 1-D1W$
- In R1 vs. R2, what is the probability that R1 wins primary? $R1 \text{ Electability} = R1W$
 $R2 \text{ Electability} = R2W = 1-R1W$
- In D1 vs. R1, what is the probability D1 wins? $D1 \text{ vs } R1 \text{ Electability} = D1R1$
 $R1 \text{ vs } D1 \text{ Electability} = R1D1 = 1-D1R1$
- In D1 vs. R2, what is the probability D1 wins? $D1 \text{ vs } R2 \text{ Electability} = D1R2$
 $R2 \text{ vs } D1 \text{ Electability} = R2D1 = 1-D1R2$
- In D2 vs. R1, what is the probability D2 wins? $D2 \text{ vs } R1 \text{ Electability} = D2R1$
 $R1 \text{ vs } D2 \text{ Electability} = R1D2 = 1-D2R1$
- In D2 vs. R2, what is the probability D2 wins? $D2 \text{ vs } R2 \text{ Electability} = D2R2$
 $R2 \text{ vs } D2 \text{ Electability} = R2D2 = 1-D2R2$

For the two candidates from your party, you calculate their "Power Number" as follows. The four columns are:

1. [Strength of Candidate]
x [Probability that opposing Candidate #1 wins primary]
x [Probability of defeating opposing Candidate #1 in General Election]
2. [Strength of Candidate]
x [Probability that opposing Candidate #2 wins primary]
x [Probability of defeating opposing Candidate #2 in General Election]
3. [Strength of opposing Candidate #1]
x [Probability that opposing Candidate #1 wins primary]
x [Probability of defeating opposing Candidate #1 in General Election]
4. [Strength of opposing Candidate #2]
x [Probability that opposing Candidate #2 wins primary]
x [Probability of defeating opposing Candidate #2 in General Election]

	Column 1	Column 2	Column 3	Column 4
D1 =	$[D1P \times R1W \times D1R1]$	$+ [D1P \times R2W \times D1R2]$	$+ [R1P \times R1W \times R1D1]$	$+ [R2P \times R2W \times R2D1]$
D2 =	$[D2P \times R1W \times D2R1]$	$+ [D2P \times R2W \times D2R2]$	$+ [R1P \times R1W \times R1D2]$	$+ [R2P \times R2W \times R2D2]$
R1 =	$[R1P \times D1W \times R1D1]$	$+ [R1P \times D2W \times R1D2]$	$+ [D1P \times D1W \times D1R1]$	$+ [D2P \times D2W \times D2R1]$
R2 =	$[R2P \times D1W \times R2D1]$	$+ [R2P \times D2W \times R2D2]$	$+ [D1P \times D1W \times D1R2]$	$+ [D2P \times D2W \times D2R2]$

Vote for the candidate from your party with the higher Power Number. If they are equal, it's a tossup.

On the next page we'll give an example.

Example - We'll assume you are a Democrat (so #6 is crossed out), with made-up numbers for the candidates. We'll assume D1 and D2 are Biden and Sanders, and R1 and R2 are Trump and DeSantis. We'll put in numbers as if you really like Sanders, but are worried he'd lose in the general election, and that you really don't like Trump, and so are considering voting for Biden instead to make sure Trump doesn't win. (As a Democrat, the odds between the two Democratic candidates don't matter here, so that line, #6 below, is crossed out.)

1. Are you a Democrat or Republican? [If Democrat, skip 6. If Republican, skip 7.]
2. On a scale of -10 to 10, how would you rate Biden as president? D1P = 5
3. On a scale of -10 to 10, how would you rate Sanders as president? D2P = 9
4. On a scale of -10 to 10, how would you rate Trump as president? R1P = -10
5. On a scale of -10 to 10, how would you rate DeSantos as president? R2P = -8
- ~~6. In Biden vs. Sanders, what is the probability that Biden wins? D1W = 0.8~~
~~D2W = 0.2~~
7. In Trump vs. DeSantos, what is the probability that Trump wins? R1W = 0.9
R2W = 0.1
8. In Biden vs. Trump, what is the probability Biden wins? D1R1 = 0.7
R1D1 = 0.3
9. In Biden vs. DeSantos, what is the probability Biden wins? D1R2 = 0.5
R2D1 = 0.5
10. In Sanders vs. Trump, what is the probability Sanders wins? D2R1 = 0.4
R1D2 = 0.6
11. In Sanders vs. DeSantos, what is the probability Sanders wins? D2R2 = 0.2
R2D2 = 0.8

Now we plug the numbers into the equation for a Democratic Primary.

$$\begin{aligned}
 D1 &= [D1P \times R1W \times D1R1] + [D1P \times R2W \times D1R2] + [R1P \times R1W \times R1D1] + [R2P \times R2W \times D2R1] \\
 &= [5 \times 0.9 \times 0.7] + [5 \times 0.1 \times 0.5] + [-10 \times 0.9 \times 0.3] + [-8 \times 0.1 \times 0.4] \\
 &= 3.15 + 0.25 + [-2.7] + [-.32] \\
 &= 0.38 \text{ for Biden}
 \end{aligned}$$

$$\begin{aligned}
 D2 &= [D2P \times R1W \times D2R1] + [D2P \times R2W \times D2R2] + [R1P \times R1W \times R1D2] + [R2P \times R2W \times R2D2] \\
 &= [9 \times 0.9 \times 0.4] + [9 \times 0.1 \times 0.2] + [-10 \times 0.9 \times 0.6] + [-8 \times 0.1 \times 0.8] \\
 &= 3.24 + 0.18 + [-5.4] + [-0.64] \\
 &= -2.62 \text{ for Sanders}
 \end{aligned}$$

Since D1 (Biden) has a power number of +0.38, while D2 (Sanders) has a power number of -2.62, you'd vote for D1 (Biden). With the numbers plugged in, Sanders is higher in Presidential, but his lower Electability numbers against Trump and DeSantis lowers his Power Number.

Here's a simpler, more practical example for how Voteometer figures out how you should vote. Suppose you are trying to decide between two Democratic primary candidates - let's say Sanders and Biden. Suppose you think Sanders would be a 10 as president, while Biden would be a 5. To keep it simple, we'll assume it's 100% that Trump wins on the Republican side, and that you think Trump would be a -10. (Remember, scale is from -10 to 10.) Let's say you figure that Sanders has only a 10% chance against Trump, while Biden has a 70% chance. Based on your inputs, Voteometer calculates the probability of each candidate winning the presidency, and multiplies that times the number you gave them for how good they'd be as president. For this, Voteometer sees two possibilities:

- If Sanders wins on the Democratic side, there's a 10% of getting a "10" president (Sanders), and a 90% chance of a "-10" president (Trump). Therefore, with Sanders, the *average* result is $(0.1 \times 10) + (0.9 \times -10) = 1 - 9 = -8$, Sanders' "Power Number."
- If Biden wins on the Democratic side, there's a 70% chance of a "5" president (Biden), and a 30% chance of a "-10" president (Trump). The *average* result is $(.7 \times 5) + (.3 \times -10) = +0.5$, Biden's "Power Number."

Since Biden's 0.5 is greater than Sanders' -8, in this case you get a better result on average by voting for Biden.

Likely Needed Expansions

1. Ability to click and choose candidates instead of hardwiring them in. However, in most elections they can be hardwired once the candidates are known. It's also not really necessary to include minor, unelectable candidates, which just clutters things up.
2. More than two candidates in a primary.
3. One candidate in opposing party (as in the example above).
4. Three or more candidates in a general election
5. Simplified version where the only evaluations are Presidential and general Electability against an average opponent from opposing party. This would take out the head-to-head Electability numbers, simplifying things.
6. Possible button that users can click that would give expert estimates of the head-to-head probabilities, so the user would only need to judge how strong each candidate would be as president, with the experts giving their judgement on electability. Many users might not have a good idea of the head-to-head chances between their candidates and opposing ones.